

Impact of Ransomware in the Derivates Market

New York University
na4101@nyu.edu

Abstract—This research paper will focus on a specific cyberattack call ransomware that are common in the financial markets, specifically the derivatives market. As traders are increasingly relying on automated software to process their trading transaction attackers are finding more loopholes within the system to attack.

Keywords— *Vulnerabilities, Security, Derivatives Market, Ransomware, cyberattack*)

I. INTRODUCTION

In recent century, we have experienced a market evolution that comes from technological innovation, with ever-so-high computing power; technology and financial market have intertwined with each other. Technology plays a pivotal role in risk management for financial derivatives product. Technology over time enabled advanced analytics tools and software applications that helps the identification and quantification of potential risks associated with derivatives trading. This new method enhances decision-making processes by providing traders and risk managers with actionable insights which is extremely valuable in making good judgements for traders. These technical platforms help traders assess exposure to various financial assets using algorithms and historical data. They help the traders identify proper hedging ratios by observing market volatility for a given underlying asset in their portfolio, essentially technology created a way to reduce undesired risk when investing that guarantees that all legal requirements are met. Unfortunately, due to the interconnected systems and massive amounts of sensitive data exchanged within these markets it ultimately exposes them to potential vulnerabilities. Which is what I will be exploring in my paper.

The organization of the paper is as follows: Section II will discuss related research regarding nature of ransomware in the financial organization (derivatives market) . In Section III, a motivating example will be discussed. Section IV will discuss a hypothetical solution of how financial organization can defend against ransomware attack. Section V will discuss the empirical evidence regarding ransomware attacks targeting trading system in financial market. Section VI will conclude the paper by discussing future work needed in the financial industry to avoid such unexpected cost.

II. RELATED RESEARCH

When a hacker deploys a patch within a software a system and blackmails in exchange of money, this is our common case of ransomware. The term "ransomware" [3] refers to the ransom letter that demands payment from the victim to restore access to their device or data, or in order to prevent the attacker from disclosing any private or embarrassing information about them. Usually, "malicious e-mail attachments, compromised websites, infected external storage devices, and corrupted software applications are how it spreads"[1]. Ransomware exposes itself during its execution to deliver the ransom demand to its victim, in contrast to other malware kinds that usually attempt to stay undiscovered. This request is typically accompanied with a message that highlights the threat and shows on the screen either before or after the encryption happens, along with a comprehensive set of instructions for making the payment, typically through a cryptocurrency. This "cybersecurity threats can lead to significant financial losses and reputational damage"[2]. The integration of advanced technologies, while beneficial for derivative trading strategies, makes these institutions prime targets for malicious activities.

III. MOTIVATING EXAMPLE

The motivation behind focusing on ransomware attack in the Derivatives market is the fact that there is little time and cost allocated to detect potential vulnerabilities within a financial organization. Team often focuses on enabling a new product, deploying their code to release. What has been happening lately is that we see rising cases of vulnerabilities as we increase our code in codebase with new technology. There must be a tracker in place to make sure data flow, codebase, and traders' analytics tool are safeguarded in a way where a hacker may never get access to the corporate environment.

IV. PROBLEM DOMAIN AND HYPOTHESIS

Ransomware can be costly and an unforeseen expense. An organization may incur additional expenses, some of which may be evident and others of which may not, in addition to the loss of revenue. Costs that are more readily apparent include: the price of paying the ransom. Then of course the price of fixing the problem, which may entail new software, hardware, and incident response services; insurance deductibles; legal fees and litigation; and public relations. Increases in insurance premiums, a decline in the value of one's reputation or trade brand, and the loss of intellectual property are some other, less evident expenses.

Currently IBM has an AI-based solution IBM QRadar ED, where it detects new and advanced threats in real time. Anomalies such as ransomware activity are detected by the AI tool (e.g., an unusual backup deletion or encryption procedure that starts unexpectedly and automatically terminates upon discovery). However, I rarely think that it is the ultimate solution. My theory is that we can create educational tools for employees, and it would be necessary to have more uniform equipment for every employee; otherwise, it would be incorrectly identifying ransomware threats more frequently, therefore AI analytic is not the only solution. I believe the way threat actor create their attacks is not standardized, and there is no one-size-fits-all method that defenders can use to keep their companies safe. We should therefore tailor our own model understanding the motivations and tactics in its own unique way. Once we understand our own unique needs, we can then configure our own anomalies detection internally which will protect our data assets and other internal information. I also think education for employees can be another effective method.

V. EMPIRICAL EVIDENCE

Analyzing metrics from financial companies who have utilized a configurable model alongside educational approach to their employees vs organization that did not have an educational approach nor a detective system in place. The sample size takes in financial organization dealing with derivatives product (that is 13% of all ransomwares attack each year). Bank1 -> Large size financial institution. Bank 2-> Midsize & Bank3 -> Small size banks. These financial institution uses a mix of tools and education to approach the the

Companies	Tools		
	<i>Detection system in place</i>	<i>education</i>	<i>Ransomwaatt rack/year</i>
Bank 1	yes	yes	217
Bank 2	yes	No	300
Bank 3	No	No	350

VI. CONCLUSION

On this paper I did research on ransomware attacks that takes place in institution where there is a derivative market in place. The conclusion seems to be that we can benefit from leveraging both fine-tuned tools and education for employees which will in the long run minimize the ransomware attack for large bank institutions. Using tools like Qualys, or QRadar in conjunction with employee education that is not yearly based is great way we can further this effort to secure the derivatives market.

[1] On the latter category of crypto-ransomware. ThJ. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.

[2] Editorial, “The Role of Technology in Derivatives: Transforming Financial Markets - Investology Hub,” *The Insurance Universe*, Sep. 24, 2024. <https://investologyhub.com/the-role-of-technology-in-derivatives/> (accessed Nov. 11, 2024).

[3] <https://www.crowdstrike.com/en-us/cybersecurity-101/ransomware/how-ransomware-spreads/>

[4] <https://blog.qualys.com/product-tech/2024/05/08/detect-address-the-top-10-mitre-attck-techniques-for-ransomware-using-policy-compliance>

[5] <https://www.statista.com/topics/4136/ransomware/#editorsPicks>